Abhishek Mallela

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Summary

Research scientist with 12+ years of experience in modeling and computation across various biological subfields. Expert in Python/TensorFlow/JAX with a strong background in mathematical modeling and deep learning. Eager to contribute to innovative projects in modeling and machine learning.

Education

Applied Math, PhD, Mar '22; GPA: 3.74	University of California at Davis, Davis, CA
Applied Stats and Analytics, MS, May '17; GPA: 3.90	University of Kansas Medical Center, Kansas City, KS
Math and Stats, MS, May '15; BS, Dec '11; GPA: 4.00; 3.81	University of Missouri at Kansas City, Kansas City, MO

Skills

- **Communication:** Cross-disciplinary expertise and proficiency in technical writing with 18 publications, 5 invited talks, 5 contributed talks, and 4 poster presentations
- **Programming:** Python, MATLAB, R, Julia, LaTeX, Mathematica, basic C++, and shell scripting
- **AI/ML:** Proficient at prompt engineering of Large Language Models (LLMs); completed Deep Learning, Natural Language Processing, and Foundations of Algorithms and Data Structures specializations (Coursera)
- **Data:** Pre- and post-processing of COVID-19 surveillance data across all US counties and county equivalents

Experience

- *Postdoctoral Fellow (Santiago Schnell's Lab at University of Notre Dame and Dartmouth College; Feb '25 to present)*
  - **Modeling**: Designing an optimal experimental protocol to estimate the parameters of enzyme-catalyzed experiments and developed a mathematical model of a minimal mechanism for hormesis in protein aggregation.
  - **AI/ML**: Collaborating with researchers at Harvard University on an effort involving repeated games with LLMs.
  - **Achievements**: 2 publications (1 as first author).
- *Postdoctoral Fellow (Center for Nonlinear Studies at Los Alamos National Laboratory; May '22 to Jan '25)*
  - **Modeling**: Estimated the basic reproduction number R_0 for COVID-19 in 50 US states and 280 urban areas, identifying regions at risk of high disease transmission and aiding future pandemic preparedness. *Techniques used: Bayesian inference, Adaptive MCMC algorithm, and convergence diagnostics.*
  - **Collaboration**: Contributed significantly to the collective efforts of a large consortium of COVID-19 researchers.
  - **Software**: Python (NumPy, SciPy, JAX, Pandas, Jupyter Notebook, Keras, TensorFlow, Numba); Julia
  - **Achievements**: 9 publications (3 as first author), 4 invited talks, 1 contributed talk, 1 poster presentation.
- *PhD Candidate, Graduate Teaching Assistant, and Tutor (University of California at Davis; Sep '17 to Mar '22)*
  - **Modeling**: Explored the stochastic aspects of coupled bistable systems arising from the Allee effect in ecology. Found that changing one parameter in multi-dimensional systems can result in tipping point cascades, common in many fields and across spatial scales. *Techniques used: Stochastic Differential Equations, Individual-Based Models, Fourier transforms, Partially Observable Markov Decision Processes*
  - **Achievements**: PhD degree with 4 first-author publications, TA and tutor for several math courses.
- *Graduate Research Associate (University of Kansas; Oct '15 to Sep '17)*
  - **Modeling**: Developed mathematical models to analyze a) post-translational modification cycles subject to synthesis and degradation, and b) length control of bacterial structure assembly. Used an agent-based framework based on the stochastic Doob-Gillespie algorithm to efficiently simulate the models numerically on a computer cluster. Achieved a 100x speedup by switching from MATLAB to Python and then C++.
  - **Software**: MATLAB, Python, C++, Kappa, BNGL, R, Mathematica
  - **Achievements**: 2 publications (1 as first author), 2 poster presentations.
- *Graduate Student Researcher and Course Instructor (University of Missouri at Kansas City; Aug '13 to May '15)*
  - **Modeling**: Designed, analyzed, and simulated an ODE model of HIV-TB co-infection. Formulated and solved an optimal control problem to simultaneously minimize disease burden and implementation cost.
  - **Collaboration**: Led project with research supervisor and external collaborator, a world-class expert in optimal control theory, to design an optimal treatment protocol for HIV-TB co-infected populations.
  - **Achievements**: 1 first-author publication, 1 invited talk, 4 contributed talks, 1 first-place poster presentation prize (500 USD), Adjunct Instructor of Record (Trigonometry and College Algebra).

RESEARCH OUTPUT

Publications (18)

- *Strategies of cooperation and defection in five large language models.* Pal, S., **Mallela, A.**, Hilbe, C., Pracher, L., Wei, C., Fu, F., Schnell, S., and Nowak, M. ('26), aRχiv preprint, Under Review
- *Structural hormesis in protein aggregation: A minimal mechanistic model.* **Mallela, A.** and Schnell, S. ('26), Journal of the Royal Society Interface, 23(237):20251037
- *Using PyBioNetFit to Leverage Qualitative and Quantitative Data in Biological Model Parameterization and Uncertainty Quantification.* Miller, E.F., **Mallela, A.**, Neumann, J., Lin, Y.T., Hlavacek, W.S., and Posner, R.G. ('25), aRχiv preprint, Accepted
- *Evaluation of FluSight influenza forecasting in the 2021-22 and 2022-23 seasons with a new target laboratory-confirmed influenza hospitalizations*. Mathis, S.M., Webber, A.E., Basu, A., **et al.** ('24), Nature Communications, 15:6289
- *Impacts of vaccination and Severe Acute Respiratory Syndrome Coronavirus 2 variants Alpha and Delta on Coronavirus Disease 2019 transmission dynamics in four metropolitan areas of the United States*. **Mallela, A.**, Chen, Y., Lin, Y.T., Miller, E.F., Neumann, J., He, Z., Nelson, K.E., Posner, R.G., and Hlavacek, W.S. ('24), Bulletin of Mathematical Biology, 86(3):31
- *Differential contagiousness of respiratory disease across the United States*. **Mallela, A.**, Lin, Y.T., and Hlavacek, W.S. ('23), Epidemics, 45:100718
- *Quantification of early nonpharmaceutical interventions aimed at slowing transmission of Coronavirus Disease 2019 in the Navajo Nation and surrounding states*. Miller, E.F., Neumann, J. Chen, Y., **Mallela, A.**, Lin, Y.T., Hlavacek, W.S., and Posner, R.G. ('23), PLOS Global Public Health, 3(6):e0001490
- *The United States COVID-19 Forecast Hub dataset*. Cramer, E.Y., Huang, Y., Wang, Y., **et al.** ('22), Scientific Data, 9(462):1-15
- *Optimal management of stochastic invasion in a metapopulation with Allee effects*. **Mallela, A.** and Hastings, A. ('22), Journal of Theoretical Biology, 549:111221
- *Bayesian inference of state-level COVID-19 basic reproduction numbers across the United States*. **Mallela, A.**, Neumann, J., Miller, E.F., Chen, Y., Posner, R.G., Lin, Y.T., and Hlavacek, W.S. ('22), Viruses, 14(1):157
- *Implementation of a practical Markov chain Monte Carlo sampling algorithm in PyBioNetFit*. Neumann, J., Lin, Y.T., **Mallela, A.**, Miller, E.F., Colvin, J., Duprat, A.T., Chen, Y., Hlavacek, W.S., and Posner, R.G. ('22), Bioinformatics, 38(6):1770-1772
- *Tipping cascades in a multi-patch system with noise and spatial coupling*. **Mallela, A.** and Hastings, A. ('21), Bulletin of Mathematical Biology, 83(11):1-27
- *Robustness and the evolution of length control strategies in the T3SS and flagellar hook*. Nariya, M.K., **Mallela, A.**, Shi, J.J., and Deeds, E.J. ('21), Biophysical Journal, 120(17):3820-3830
- *The role of stochasticity in noise-induced tipping cascades: A master equation approach*. **Mallela, A.** and Hastings, A. ('21), Bulletin of Mathematical Biology, 83(5):1-20
- *Daily forecasting of regional epidemics of Coronavirus disease with Bayesian uncertainty quantification*. Lin, Y.T., Neumann, J., Miller, E.F., Posner, R.G., **Mallela, A.**, Safta, C., Ray, J., Thakur, G., Chinthavali, S., and Hlavacek, W.S. ('21), Emerging Infectious Diseases, 27(3):767
- *Crosstalk and ultrasensitivity in protein degradation pathways*. **Mallela, A.**, Nariya, M.K., and Deeds, E.J. ('20), PLOS Computational Biology, 16(12):e1008492
- *Optimal control applied to a SEIR model of 2019-nCoV with social distancing*. **Mallela, A.** ('20), medRχiv preprint
- *HIV-TB co-infection treatment: modeling and optimal control theory perspectives*. **Mallela, A.**, Lenhart, S., and Vaidya, N.K. ('16), Journal of Computational and Applied Mathematics, 307:143-161

Invited talks (5)

- Joint Mathematics Meetings, San Francisco, CA (Jan '24)
  - *Differential contagiousness of respiratory disease across the United States*
- CNLS at LANL, Los Alamos, NM
  - *Impacts of vaccination and SARS-CoV-2 variants Alpha and Delta on COVID-19 transmission dynamics in four metropolitan areas of the US* (Jun '23)
  - *Differential contagiousness of respiratory disease across the United States* (Sep '22)
  - *Crosstalk and ultrasensitivity in protein degradation pathways* (Dec '21)
- SIAM Central States Sectional Meeting, Rolla, MO (Apr '15)
  - *Ideal treatments for HIV-TB co-infected populations: modeling & optimal control theory perspectives*

Contributed talks (5) & poster presentations (4)

- SIAM Conference on Uncertainty Quantification, Trieste, Italy (Mar '24)
  - *Talk: Differential contagiousness of respiratory disease across the United States*
- Quantitative and Systems Biology (q-bio) conference
  - *Poster: Bayesian inference with PyBioNetFit of state-level R_0 values for COVID-19 across the US;* Fort Collins, CO (Jun '22)
  - *Poster: Protein turnover impacts dynamics of post-translational modifications;* Nashville, TN (Jul '16)
- Conference on Modeling Protein-Protein Interactions, Lawrence, KS (Oct '16)
  - *Poster: Protein turnover impacts dynamics of post-translational modifications*
- SIAM Conference on Applications of Dynamical Systems, Snowbird, UT (May '15)
  - *Talk: Optimal treatment strategies for HIV-TB co-infected individuals*
- UMKC Mathematics and Statistics Research Day, Kansas City, MO (Apr '15)
  - *Talk: Ideal treatments for HIV-TB co-infected populations: modeling and optimal control theory perspectives*
- UMKC Community of Scholars Symposium, Kansas City, MO
  - *Talk: Optimal treatment strategies for HIV-TB co-infected populations* (Apr '15)
  - *Poster: Optimal treatment strategies for HIV-TB co-infected populations* (May '14)
- Joint Mathematics Meetings, San Antonio, TX (Jan '15)

- *Talk: Optimal treatment strategies for HIV-TB co-infected populations*